

Original LCTRSs (3 Processes)

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times Loc \times int \times int \times int \times int \times int \Rightarrow State \\ \text{noncrit} : Loc \\ \text{level1} : Loc \\ \text{level2} : Loc \\ \text{crit} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\Gamma \left\{ \begin{array}{ll} 1 : & \text{state}(\text{noncrit}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{level1}, \text{proc1}, \text{proc2}, p01, p1, p2, y11, y2) \quad [p01 = 1 \wedge y11 = 0] \\ 2 : & \text{state}(\text{level1}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{level2}, \text{proc1}, \text{proc2}, p01, p1, p2, y1, y21) \quad [(y1 \neq 0 \vee ((p1 < 1) \wedge (p2 < 1))) \wedge p01 = 2 \wedge y21 = 0] \\ 3 : & \text{state}(\text{level2}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{crit}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \quad [y2 \neq 0 \vee ((p1 < 2) \wedge (p2 < 2))] \\ 4 : & \text{state}(\text{crit}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{noncrit}, \text{proc1}, \text{proc2}, p01, p1, p2, y1, y2) \quad [p01 = 0] \\ 5 : & \text{state}(\text{proc0}, \text{noncrit}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{level1}, \text{proc2}, p0, p11, p2, y11, y2) \quad [p11 = 1 \wedge y11 = 1] \\ 6 : & \text{state}(\text{proc0}, \text{level1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{level2}, \text{proc2}, p0, p11, p2, y1, y21) \quad [(y1 \neq 1 \vee ((p0 < 1) \wedge (p2 < 1))) \wedge p11 = 2 \wedge y21 = 1] \\ 7 : & \text{state}(\text{proc0}, \text{level2}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{crit}, \text{proc2}, p0, p1, p2, y1, y2) \quad [y2 \neq 1 \vee ((p0 < 2) \wedge (p2 < 2))] \\ 8 : & \text{state}(\text{proc0}, \text{crit}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{noncrit}, \text{proc2}, p0, p11, p2, y1, y2) \quad [p11 = 0] \\ 9 : & \text{state}(\text{proc0}, \text{proc1}, \text{noncrit}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{proc1}, \text{level1}, p0, p1, p21, y11, y2) \quad [p21 = 1 \wedge y11 = 2] \\ 10 : & \text{state}(\text{proc0}, \text{proc1}, \text{level1}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{proc1}, \text{level2}, p0, p1, p21, y1, y21) \quad [(y1 \neq 2 \vee ((p0 < 1) \wedge (p1 < 1))) \wedge p21 = 2 \wedge y21 = 2] \\ 11 : & \text{state}(\text{proc0}, \text{proc1}, \text{level2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{proc1}, \text{crit}, p0, p1, p2, y1, y2) \quad [y2 \neq 2 \vee ((p0 < 2) \wedge (p1 < 2))] \\ 12 : & \text{state}(\text{proc0}, \text{proc1}, \text{crit}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{proc1}, \text{noncrit}, p0, p1, p21, y1, y2) \quad [p21 = 0] \\ 13 : & \text{state}(\text{proc0}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{success} \\ 14 : & \text{state}(\text{crit}, \text{crit}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{error} \\ 15 : & \text{state}(\text{crit}, \text{proc1}, \text{crit}, p0, p1, p2, y1, y2) \rightarrow \text{error} \\ 16 : & \text{state}(\text{proc0}, \text{crit}, \text{crit}, p0, p1, p2, y1, y2) \rightarrow \text{error} \end{array} \right\}$$

APR Problem

$$\{ \quad 17 : \quad \text{state}(\text{noncrit}, \text{noncrit}, \text{noncrit}, 0, 0, 0, y1, y2) \Rightarrow \text{success} \quad \}$$

BV-LCTRS Version (3 Processes)

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times Loc \times bitvec_2 \times bitvec_2 \times bitvec_2 \times bitvec_2 \times bitvec_2 \Rightarrow State \\ \text{noncrit} : Loc \\ \text{level1} : Loc \\ \text{level2} : Loc \\ \text{crit} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$2 \left\{ \begin{array}{ll} 1 : & \text{state}(\text{noncrit}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{level1}, \text{proc1}, \text{proc2}, p01, p1, p2, y11, y2) \quad [p01 = \#b01 \wedge y11 = \#b00] \\ 2 : & \text{state}(\text{level1}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{level2}, \text{proc1}, \text{proc2}, p01, p1, p2, y1, y21) \quad [(y1 \neq \#b00 \vee ((p1\text{bvult}\#b01) \wedge (p2\text{bvult}\#b01))) \wedge p01 = \#b10 \wedge y21 = \#b00] \\ 3 : & \text{state}(\text{level2}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{crit}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \quad [y2 \neq \#b00 \vee ((p1\text{bvult}\#b10) \wedge (p2\text{bvult}\#b10))] \\ 4 : & \text{state}(\text{crit}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{noncrit}, \text{proc1}, \text{proc2}, p01, p1, p2, y1, y2) \quad [p01 = \#b00] \\ 5 : & \text{state}(\text{proc0}, \text{noncrit}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{level1}, \text{proc2}, p0, p11, p2, y11, y2) \quad [p11 = \#b01 \wedge y11 = \#b01] \\ 6 : & \text{state}(\text{proc0}, \text{level1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{level2}, \text{proc2}, p0, p11, p2, y1, y21) \quad [(y1 \neq \#b01 \vee ((p0\text{bvult}\#b01) \wedge (p2\text{bvult}\#b01))) \wedge p11 = \#b10 \wedge y21 = \#b01] \\ 7 : & \text{state}(\text{proc0}, \text{level2}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{crit}, \text{proc2}, p0, p1, p2, y1, y2) \quad [y2 \neq \#b01 \vee ((p0\text{bvult}\#b10) \wedge (p2\text{bvult}\#b10))] \\ 8 : & \text{state}(\text{proc0}, \text{crit}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{noncrit}, \text{proc2}, p0, p11, p2, y1, y2) \quad [p11 = \#b00] \\ 9 : & \text{state}(\text{proc0}, \text{proc1}, \text{noncrit}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{proc1}, \text{level1}, p0, p1, p21, y11, y2) \quad [p21 = \#b01 \wedge y11 = \#b10] \\ 10 : & \text{state}(\text{proc0}, \text{proc1}, \text{level1}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{proc1}, \text{level2}, p0, p1, p21, y1, y21) \quad [(y1 \neq \#b10 \vee ((p0\text{bvult}\#b01) \wedge (p1\text{bvult}\#b01))) \wedge p21 = \#b10 \wedge y21 = \#b10] \\ 11 : & \text{state}(\text{proc0}, \text{proc1}, \text{level2}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{proc1}, \text{crit}, p0, p1, p2, y1, y2) \quad [y2 \neq \#b10 \vee ((p0\text{bvult}\#b10) \wedge (p1\text{bvult}\#b10))] \\ 12 : & \text{state}(\text{proc0}, \text{proc1}, \text{crit}, p0, p1, p2, y1, y2) \rightarrow \text{state}(\text{proc0}, \text{proc1}, \text{noncrit}, p0, p1, p21, y1, y2) \quad [p21 = \#b00] \\ 13 : & \text{state}(\text{proc0}, \text{proc1}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{success} \\ 14 : & \text{state}(\text{crit}, \text{crit}, \text{proc2}, p0, p1, p2, y1, y2) \rightarrow \text{error} \\ 15 : & \text{state}(\text{crit}, \text{proc1}, \text{crit}, p0, p1, p2, y1, y2) \rightarrow \text{error} \\ 16 : & \text{state}(\text{proc0}, \text{crit}, \text{crit}, p0, p1, p2, y1, y2) \rightarrow \text{error} \end{array} \right\}$$

APR Problem

$$\{ \quad 17 : \quad \text{state}(\text{noncrit}, \text{noncrit}, \text{noncrit}, \#b00, \#b00, \#b00, y1, y2) \Rightarrow \text{success} \quad \}$$